

A150 · Resolving Mentions to Ontology Gaps in Biomedical Entity Linking

Presenter: Hyun Seung Lim — Northeastern University

Authors: Hyun Seung Lim, Benjamin M. Gyori

Session: Day 1 · Thu Oct 1, 2026 · 2:15–4:15 PM

Recognizing names or synonyms of entities such as genes, small molecules and diseases in text and normalizing them to ontology terms is called entity linking (EL) and is a central task for biomedical data integration. As published biomedical knowledge rapidly evolves, a key challenge in EL is the incompleteness of ontologies where a given entity mention is novel or sits at a granularity no existing term occupies. While encoder-based methods such as SapBERT excel on biomedical entity linking benchmarks, they are limited to predicting exact equivalence and don't handle incompleteness in a principled way. To address this challenge, we reformulate entity linking so that a mention resolves either to an existing term or, when it is unlinkable, to the region between terms that bracket it in a semantic sense. Creating the training data requires no manual annotation as the ontology's own subsumption edges supply the ordering between terms, and synonyms are obtained from the UMLS Metathesaurus. We use this training data to fine tune a SapBERT-initialized encoder which learns both semantic equivalence and subsumption relationships between two terms, allowing it to predict subsumption when two terms aren't equivalent. In an evaluation against a similarity-based linker as a control, our approach matched it on mentions carrying gold ontology terms while also abstaining on a portion of mentions the ontology cannot represent. We also present evaluation results against a public benchmark of literature mentions that includes terms absent from the ontology, making incompleteness explicit.